Engineering - Graduate Level Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Find the solution for the equationX=AX+bu;X(0) = X_0 where A=\{}vert10\{}vert,b=\{}vert1\{}v
=\{}vert- 1\{}vert, \{}vert- 22\{}vert\{}vert2\{}vert\{}vert0\{}vert u(t) =...
   (a) -3
   (b) 2
   (c) 0
   (d) 3
   (e) 10

2. A dc motor requires 10 kilowatts to enable it to supply its full capacity of 10 horsepower to its pulley. What is its full-load efficiency ?
   (a) 74.6%
   (b) 65%
   (c) 50%
   (d) 85%
   (e) 92%

3. Find the input impedance of a 50-ohm line terminated in +j50 ohms, for a line length such that\{}betad= (\{}pi/1) radian.
   (a) 100 ohms
   (b) 25 ohms
   (c) j100 ohms
   (d) 50 ohms
   (e) - j50 ohms

4. The number of rings in the Bohr model of any element is determined by what?
   (a) Number of isotopes.
   (b) Atomic number.
   (c) The element's period on the periodic table.
   (d) Column number on periodic table.
   (e) Atomic radius.

5. The errors mainly caused by human mistakes are
   (a) gross error.
   (b) instrumental error.
   (c) systematic error.

True / False (10 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Which of these sets of logic gates are designated as universal gates?' is 'OR, XOR, NOT.'.

2. True or False: The correct answer to 'What is steady state heat transfer?' is 'Steady state heat transfer is when the rate of energy in equals the rate of energy out, with no accumulation of energy.'.

3. True or False: The correct answer to 'Consider a capacitor with capacitance C = 10ˆ-6 farad. Assume that initial voltage across this capacitor is\{}nu_c(0) = 1 volt. Find the...' is 'sin (10ˆ6 t) - 1'.

4. True or False: The correct answer to 'Use Laplace's equation to find the capacitance of a parallel platecapacitor.' is 'C= (\{}epsilonS/ d)'.

5. True or False: The correct answer to 'The ROM programmed during manufacturing process itself is called' is 'MROM'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. A cast-iron pulley rotates on a motor shaft at 250 rpm and delivers 50 h.p. Calculate the pulley diameter if density (\{}rho) of cast-iron is 0.05 lb/inˆ3 and tensile stress is 150 lbs/in?

2. Find the inverse Laplace transform Lˆ-1[1 / {s(sˆ2 + 4)}], using the convolution.

End of Paper